

Stefano Racioppi, Ph.D.

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Professional Experience

- 09/2025 – present Research Associate, *University of Cambridge*, UK.
Group: Prof. Chris Pickard.
- 03/2021 – 08/2025 Postdoctoral researcher, *State University of New York at Buffalo*, USA.
Group: Prof. Eva Zurek.
- 07/2023 – 09/2023 Visitor, *Aalto University*, Finland.
Group: Prof. Patrick Rinke.
- 02/2019 – 02/2021 Postdoctoral researcher, *Chalmers University of Technology*, Sweden.
Group: Prof. Martin Rahm.



Education

- 10/2015 – 01/2019 Ph.D. in Industrial Chemistry, *University of Milan*, Italy. Supervisors: Prof. Angelo Sironi; Prof. Piero Macchi; Prof. Pierluigi Mercandelli.
- 11/2018 – 01/2019 Scholarship recipient, National Interuniversity Consortium of Materials Science and Technology (INSTM), *University of Milan*, Italy. Supervisor: Prof. Angelo Sironi.
- 10/2016 – 03/2018 Visiting student at the *University of Bern*, Switzerland. Host: Prof. Piero Macchi.
- 10/2013 – 10/2015 Master's degree in Industrial Chemistry, *University of Padua*, Italy. Supervisors: Prof. Armando Gennaro; Prof. Laura Orian. Final grade: 110/110
- 10/2010 – 09/2013 Bachelor's degree in Industrial Chemistry, *University of Padua*, Italy. Supervisor: Prof. Armando Gennaro.

Research Interests

- Computational and Theoretical Chemistry.
- Quantum Crystallography and Crystal Structure Prediction.
- High Pressure Chemistry.
- Chemical Bonding.
- Materials and Planetary Science.
- Superconductivity.
- Machine-Learning.

Distinctions

- 02/2026 Marie Skłodowska-Curie Actions (MSCA) Postdoctoral Fellowship, Seal of Excellence 2026.
- 01/2026 Research Development Fellows Program – University of Cambridge, Centre for Teaching and Learning.
- 03/2025 Marie Skłodowska-Curie Actions (MSCA) Postdoctoral Fellowship, Seal of Excellence 2025.
- 09/2019 Best Ph.D. Thesis Award by the Italian Crystallographic Association (AIC).
- 07/2017 Outstanding poster prize offered by Wiley – Journal of Computational Chemistry and International Journal of Quantum Chemistry. Workshop on ab Initio Modeling in Solid State Chemistry with CRYSTAL, *Minneapolis*, USA.

Memberships

- 2026 – present European Committee for Chemical Bonding.
- 2025 – present Trinity Hall College of Cambridge.
- 2024 – present European Crystallographic Association (ECA).
- 2021 – present American Physical Society (APS).
- 2018 – present Italian Crystallographic Association (AIC).
- 2018 – 2019 Swiss Society for Crystallographic (SGK/SSCr).

External Fundings, Proposals and Scholarships

05/2025	Awarded: Proposal experiment, <i>Structural Determination of Al₂O₃ at super-Earth Conditions – AIO Discovery</i> , Omega Laser Facility at the Laboratory for Laser Energetics (LLE); 1 shot day awarded. PI: Dr. Racioppi S.
04/2025	Awarded: Beamtime at the European Synchrotron Radiation Facility (ESRF); <i>Diagnosing Electride Formation in Ca up to 40 GPa with XANES and SCXRD</i> ; 18 shifts awarded. PI: Dr. Chin, D. A.; co-PI: Dr. Racioppi, S.
04/2024	Awarded: 34 th European Crystallographic Meeting Scholarship; € 300.
09/2023 – 08/2026	Contributed to the proposal's writing: <i>High Energy Density Quantum Matter - Renewal</i> ; Award number: LOI-0000043230; Source of Support: Department of Energy (DOE); PI: Dr. Collins G.W.; Co-PIs: Dr. Deemyad S.; Dr. Hemley R. J.; Dr. Zurek E.; \$ 2,100,000.
02/2023	Awarded: Beamtime at Laser-lab Europe LENS; <i>Tuning Electrical Conductivity of Aurophilic Interactions by Compression</i> (ID: 22396); PIs: Dr. Racioppi S., Dr. Poreba T.; Prof. Macchi P.
09/2022	Awarded: Seed Funding Project; Center for Matter at Atomic Pressure (CMAP); \$ 3500.
10/2018 – 01/2019	Awarded: Scholarship; National Interuniversity Consortium of Materials Science and Technology (INSTM); PI: Prof. Sironi A.; € 5862.
05/2018	Awarded: Travelling Grant; Ettore Majorana Foundation; € 600.

Publications

Total number of publications: 26 (16 as first author, 6 as corresponding author (*)).

Total number of citations: 290.

H-index: 10.

- (31.) **Racioppi, S.**; Kelly, A.; Zurek, E. Computational Investigation of Al₂O₃ Structures at Super-Earth Interior Conditions, *submitted*.
- (30.) Liu, Z.; **Racioppi, S.**; Hilleke, K. P.; Pandit, A.; Ma, S.; Hermann, A.; Yan, D.; Zurek, E.; Miao, M. Non-local Chemistry Driven by Cation-Anion Size Disparity in Helium Inserted Compounds under High Pressure, *arXiv:2511.23400*.
- (29.) Farraj, H.; **Racioppi, S.**; Garbarino, G.; Ahart, M.; Mondal, A.; Parra, S. G.; Smith, J. S.; Cohen, R. E.; Zurek, E.; Cabana, J.; Hemley, R. Novel Pressure-Induced Transformations of PbTiO₃, *arXiv:2511.05756*.
- (28.) **Racioppi, S.***; Zurek, E. Quantum Effects or Theoretical Artifacts? A Computational Reanalysis of Hydrogen at High-Pressure, *arXiv:2510.02098*.
- (27.) Ocampo, I. K.; Kim, D.; Smith, R. F.; Coppari, F.; **Racioppi, S.**; Zurek, E.; Rygg, J. R.; Chin, D. A.; Millot, M.; Eggert, J. H.; Duffy, T. S.; Atomic-level structure and pressure-density response of Fe-O compounds at deep Earth and exoplanetary interior conditions, *under review in Physics of the Earth and Planetary Interiors*.
26. Weadock, N.; Holle, W.; Myford, K.; **Racioppi, S.**; Ghimire, A.; Ladd, D.; Park, C.; Zurek, E.; Toney, M.; Bridges, F.; Deemyad, S. Pressure-driven helium insertion for structural stability of CH₃NH₃PbBr₃ hybrid perovskites, *Chem. Mater.*, **2026**.
25. Caussé, M.; Bozier, K.; Cooke, P. I. C.; **Racioppi, S.**; Pickard C. J. Metastability and high-T_c superconductivity in Al₅-type ternary hydride YSbH₆ at moderate pressure, *PRB*, **2026**.
24. Coleman, A. L.; Singh, S.; Lockard, T.; Ocampo, I. K.; Lazicki, A. E.; Gorman, M. G.; **Racioppi, S.**; Krygier, A. G.; Wehrenberg, C. E.; Ahmad, R.; Hamel, S.; Han, S.; Ginnane, M. K.; Swift, D. C.; Bonev, S. A.; Zurek, E.; Duffy, T. S.; Eggert, J. H.; McNaney, J.; Smith, R. F. Body-centered cubic phase transformation in gold at TPa pressures. *PRL*, **2025**.
23. **Racioppi, S.***; Zurek, E. Using Topology to Predict Electrides in the Solid, *J. Phys. Chem. A*, **2025**.
22. **Racioppi, S.***; Zurek, E. Activation of Semicore Electrons in Alkali Metals and Their Role in the B1–B2 Phase Transition under Pressure, *J. Am. Chem. Soc.*, **2025**, 147, 36, 32745-32751.
21. Storm, C. V.; **Racioppi, S.**; Duff, M. J.; McHardy, J. D.; Zurek E.; McMahon, M. I. Experimental Evidence of Interstitial Electron Density in Transparent Dense Sodium, *Commun. Mater.*, **2025**, 6, 201.
20. **Racioppi, S.**; Zurek, E. High-Pressure Electrides: A Quantum Chemical Perspective, *Annu. Rev. Mater. Res.*, **2025**, 55, 421-442.
19. **Racioppi, S.**; Saffarin-Deemyad, I.; Holle, W.; Belli, F.; Ferry, R.; Kenney-Benson, C.; Smith, C. J.; Zurek, E.; Deemyad, S. Phase Lithium's low-temperature phase transitions: Insights into quantum lattice dynamics and superconductivity, *PRB*, **2025**, 111, 054111.
18. **Racioppi, S.**; Zurek, E. Looking at High-Pressure Electrides Through the Lens of Quantum Crystallography: The Case of Simple Cubic Calcium, *Acta Cryst.*, **2025**, B81.
17. **Racioppi, S.**; De la Roza, A.; Hajinazar, S.; Zurek, E. Powder X-Ray Diffraction Assisted Evolutionary Algorithm for Crystal Structure Prediction, *Digit. Discov.*, **2025**, 4, 73-83.

16. **Racioppi, S.**; De la Roza, A.; Hajinazar, S.; Zurek, E. PXRD-Assisted Crystal Structure Predictions, *Acta Cryst.*, **2024**, A80, e396.
15. **Racioppi, S.**; P.; Hyldgaard, P.; Rahm, M. Quantifying Atomic Volume, Partial Charge and Electronegativity in Condensed Phases, *J. Phys. Chem. C*, **2024**, 128, 4009-4017.
14. **Racioppi, S.***; Storm, C. V.; McMahon, M. I.; Zurek, E. On the Electride Nature of Na-hP4, *Angew. Chem. Int. Ed.* **2023**, e202310802.
13. **Racioppi, S.**; Miao, M.; Zurek, E. Intercalating Helium into A-site Vacant Perovskites, *Chem. Mater.* **2023**, 35, 4297-4310.
12. Fang, M.; Srikanth Kumar, G.; **Racioppi, S.**; Zhang, H.; Zurek, E.; Lin, Q. Hydrazonyl Sultones as Stable Tautomers of Highly Reactive Nitrile Imines for Fast Bioorthogonal Ligation Reaction, *J. Am. Chem. Soc.* **2023**, 145, 18, 9959-9964.
11. **Racioppi, S.**; Lolur, P.; Hyldgaard, P.; Rahm, M. A Density Functional Theory for the Average Electron Energy, *J. Chem. Theory Comput.* **2023**, 19, 799-807.
10. Antle, J.; Kimura, M.; **Racioppi, S.**; Lang, M.; Damon, C.; Gatley-Montross, C.; Sánchez B., L.; Miller, D.; Zurek, E.; Brown, A.; Gast, K.; Simpson, S. Applying Density Functional Theory to Common Organic Mechanisms: A Computational Exercise, *J. Chem. Educ.* **2023**, 100, 355-360.
9. **Racioppi, S.**; Orian, L.; Gennaro, A.; Isse, A. Solvent Coordination Effect on Copper-Based Molecular Catalysts for Controlled Radical Polymerization, *Catalysts* **2022**, 12, 1656.
8. Poręba, T.; **Racioppi, S.***; Garbarino, G.; Morgenroth, W.; Mezouar, M. Investigating the Structural Symmetrization of CsI₃ at High Pressures through Combined X-ray Diffraction Experiments and Theoretical Analysis, *Inorg. Chem.* **2022**, 61, 10977-10985.
7. Della Pergola, R.; Garlaschelli, L.; Macchi, P.; Ruffo, R.; **Racioppi, S.**; Sironi, A. From Small Metal Clusters to Molecular Nanoarchitectures with Core-Shell Structure: Synthesis, Redox Fingerprint, Theoretical Analysis and Solid-State Structure of [Co₃₈As₁₂(CO)₅₀]⁴⁺, *Inorg. Chem.* **2022**, 61, 9888-9896.
6. Kumar, G. S.; **Racioppi, S.**; Zurek, E.; Lin, Q. Superfast Tetrazole-BCN Cycloaddition Reaction for Bioorthogonal Protein Labeling on Live Cells, *J. Am. Chem. Soc.* **2022**, 144, 57-62.
5. **Racioppi, S.**; Rahm M. In-Situ Electronegativity and the Bridging of Chemical Bonding Concepts, *Chem. Eur. J.* **2021**, 27, 18156-18167.
4. **Racioppi, S.***; Sironi, A.; Macchi, P. On Generalized Partition Methods for Interaction Energies, *Phys. Chem. Chem. Phys.* **2020**, 24291-24298.
3. **Racioppi, S.**; Andrzejewski, M.; Colombo, V.; Sironi, A.; Macchi, P. Different Metallophilic Attitudes Revealed by Compression, *Inorg. Chem.* **2020**, 59, 2223-2227.
2. **Racioppi, S.***; Della Pergola, R.; Colombo V.; Sironi, A.; Macchi, P. Electron Density Analysis of Metal Clusters with Semi-Interstitial Main Group Atoms. Chemical Bonding in [Co₆X(CO)₁₆]⁻ Species, *J. Phys. Chem. A* **2018**, 122, 5004-5015.
1. Della Pergola, R.; Sironi, A.; Colombo, V.; Garlaschelli, L.; **Racioppi, S.**; Sironi, A.; Macchi, P. Periodical trends in [Co₆E(CO)₁₆]⁻ clusters: Structural, synthetic and energy changes produced by substitution of P with As, *J. Organomet. Chem.* **2017**, 849-850, 130-136.

Invited and Contributed Talks and Seminars

Talks

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| 08/2026 | 27 th Congress and General Assembly of the International Union of Crystallography; Microsymposium MS084: Quantum crystallography of non-equilibrium or nanostructured materials.* |
| 06/2025 | 24 th Biennial International Conference on the Science of Compression in Condensed Matter (SCCM): Computational Methods: Theory, Codes and Applications (FMDA/B). |
| 06/2025 | Crystal Structure Prediction Workshop, a Practical Approach, Poitiers. |
| 09/2024 | 4 th European Symposium on Chemical Bonding (CBOND2024). |
| 08/2024 | 34 th European Crystallographic Meeting (ECM34); Microsymposium M27: High Pressure Crystallography: Exploring Structure and Method Development at Extreme Conditions. |
| 07/2024 | Gordon Research Seminar (GRS), Research at High Pressure.* |
| 07/2023 | 28 th AIRAPT and 60 th EHPRG International Conference on High Pressure Science and Technology; Session: Chemical Bonding. |
| 03/2022 | American Physical Society (APS) March Meeting; Session G24: Matter at Extreme Conditions: Phase Transitions. |
| 09/2021 | Italian Crystallographic Association (AIC) XLIX Meeting.* |
| 06/2018 | International School of Crystallography in Erice; 52 nd Course: Quantum Crystallography. |

Seminars

- 04/2026 *Diamond's Science Happy Hour*, Diamond Light Source Ltd; *Hosts*: Dr. Annette Kleppe.*
- 03/2026 Italian Young Crystallographers (GCI) *GCI Insights* Virtual Seminar; *Host*: Dr. Martina Lippi.*
- 11/2025 Chicago/DOE Alliance Center (CDA) Virtual Seminar Series; *Host*: Prof. Russell Hemley.*
- 05/2025 Cambridge Crystallographic Data Centre (CCDC), Virtual Seminar; *Host*: Dr. Fabio Montisci.*
- 07/2024 High Energy Density Science Center (HEDS) Virtual Seminar Series; Lawrence Livermore National Laboratory (LLNL, USA); *Host*: Dr. Federica Coppari.*
- 01/2024 *University of Milan* (Italy), Dep. of Chemistry; *Host*: Prof. Davide Proserpio.*
- 08/2023 *Aalto University* (Finland), Dep. of Physics; *Host*: Prof. Patrick Rinke.*
- 06/2023 *University of Oviedo* (Spain), Dep. of Chemistry; *Host*: Prof. José Manuel Recio.*
- 04/2023 Italian Young Crystallographers (GCI) *Happy Hour* Virtual Seminar; *Host*: Dr. Marta Morana.*
- 03/2022 Chicago/DOE Alliance Center (CDA) Virtual Seminar Series; *Host*: Prof. Russell Hemley.*

* = invited

Appointments and Services

- 07/2026 Elected chair of the Gordon Research Seminar (GRS), Research at High Pressure.
- 07/2026 Co-organizer of the “*Under Pressure: from Chemical Bonding to Planetary Structures and Evolution*” session at the Goldschmidt 2026 conference.
- 03/2026 Appointed Springer Nature Reviewing Editor of Nature Portfolio journals.
- 07/2025 Co-organizer of the “*High Energy Density Physics: Quantum Materials and Quantum Phenomena*” session at the 24th Biennial Conference on Science of Compression in Condensed Matter (SCCM).
- 06/2025 Co-organizer of a workshop on the crystal structure prediction code XtalOpt (co-organizer: Prof. Eva Zurek) at the Crystal Structure Prediction Workshop, a Practical Approach, held in Poitiers, France.
- 05/2025 Moderator at the *Graduate Student Symposium* (GSS) at SUNY Buffalo.
- 06/2022 Co-organizer of a workshop on the crystal structure prediction code XtalOpt (co-organizer: Prof. Eva Zurek) at the International School of Crystallography in Erice; 56th Course: Crystallography under extreme conditions.
- 06/2022 Co-organizer of a workshop on computational chemistry techniques (co-organizer: Dr. Julia Contreras-García) at the International School of Crystallography in Erice; 56th Course: Crystallography under extreme conditions.

Teaching and Supervision

- 2026 Supervisor of one student for the MPhil project in Scientific Computing – University of Cambridge.
- 2026 Supervisor of two students for the MPhil written assignment in Scientific Computing – University of Cambridge.
- 2025 Demonstrator for a course in the Scientific Computing MPhil on Electronic Structure and Density Functional Theory – University of Cambridge.
- 2021/2025 Co-supervisor of three PhD students (Morgan Reddington, Masashi Kimura, Isaac Ampersand-Kwadwo Akwetey) and supervisor of two undergraduate students (Amelia Kelly, Jarrod DuMond) – SUNY Buffalo.
- 01/2024 Special virtual lecture for the students of the general chemistry course at the *University of Milan* – Host: Prof. Davide Proserpio. “*History and Evolution of the Electronegativity*”
- 2022 – 2023 Hosted online workshops for the usage of the crystal structure prediction code XtalOpt.
- 2021 Supervisor of a high school student (Henry Feng) for summer project – SUNY Buffalo.
- 2017 – 2019 Co-supervisor of two master students (Stephanie Terruzzi and Simona Sorbara) – University of Milan.
- 2018 – 2019 Teaching assistant, physical chemistry II, 2nd year degree in Industrial Chemistry – University of Milan.
- 2015 – 2016 Teaching assistant, general and inorganic chemistry, 3rd year degree in Physics – University of Milan.

Computational Developments

Electronegativity in gas-phase: <https://github.com/SteRacioppi/Chi-bar-Atomic-and-Molecular-Electronegativity>
Electronegativity in solid: https://www.scm.com/doc/BAND/Expert_Options/Restarts.html

Public Service and Outreach

- 2026 *Knowable Magazine*, [Meet the mysterious electrifieds.](#)
- 2025 *News Center - News & information from UB*, [Core electron bonding may not always require extreme pressure, study finds.](#)
- 2025 *Science News*, [A new iron compound hints 'primordial' helium hides in Earth's core.](#)
- 2023 *News Center - News & information from UB*, [Sodium's high-pressure transformation can tell us about the interiors of stars, planets.](#)